

Magnitude Comparator

$A = A_3 A_2 A_1 A_0$ } Design a ckt to say
 $B = B_3 B_2 B_1 B_0$ } a) if $A > B$ b) if $A = B$
c) if $A < B$

Obviously it is easy to see that
 $A = B$ if $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$
and $A_0 = B_0$. In other words.

$A = B$ is realized by the ckt

$$(A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

where $X_i = A_i \odot B_i$ is A XNOR B_i

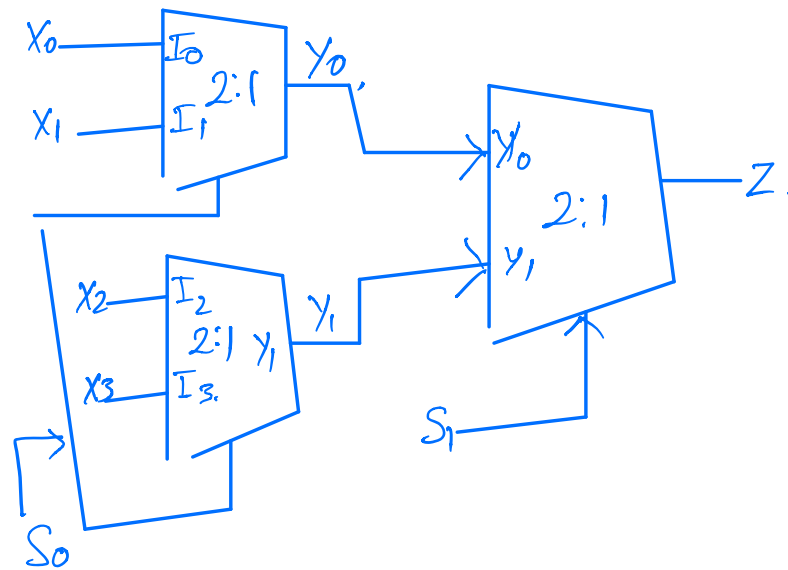
$A > B$ is realized by

$$(A_3 \bar{B}_3) + (X_3 A_2 \bar{B}_2) + (X_3 X_2 A_1 \bar{B}_1) + (X_3 X_2 X_1 A_0 \bar{B}_0)$$

$A < B$

$$(\bar{A}_3 B_3) + (X_3 \bar{A}_2 B_2) + (X_3 X_2 \bar{A}_1 B_1) + (X_3 X_2 X_1 \bar{A}_0 B_0)$$

Realize a 4:1 MUX from 2:1 MUXs.



Analytically:

$$\begin{aligned}
 Z &= \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3 \\
 &= \bar{S}_1 (\underbrace{\bar{S}_0 I_0 + S_0 I_1}_{2:1 \text{ MUX}}) + S_1 (\underbrace{\bar{S}_0 I_2 + S_0 I_3}_{2:1 \text{ MUX}}) \\
 &= \bar{S}_1 Y_0 + S_1 Y_1.
 \end{aligned}$$

Design a 4 bit 2's complementer:

x_3	x_2	x_1	x_0	C	z_3	z_2	z_1	z_0
0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	1	1	1
0	0	1	0	0	1	1	1	0
0	0	1	1	0	1	1	0	1
1	1	1	1	0	0	0	0	1

Clearly $z_0 = x_0$, $z_1 = x_1$ if $x_0 = 0$ and $\bar{x}_1$ if $x_0 = 1$

$$z_1 = x_1 \bar{x}_0 + \bar{x}_1 x_0 = x_1 \oplus x_0$$

$$\begin{aligned} z_2 &= x_2 \text{ if } (x_1 = x_0 = 0) \text{ and } \bar{x}_2 \text{ if } (x_1 \text{ OR } x_0 = 1) \\ &= x_2 (\bar{x}_1 \bar{x}_0) + \bar{x}_2 (x_1 + x_0) \\ &= x_2 (\overline{x_1 + x_0}) + \bar{x}_2 (x_1 + x_0) \end{aligned}$$

$$\Rightarrow z_2 = x_2 \oplus (x_1 + x_0)$$

$$z_3 = x_3 \text{ if } (x_2 = x_1 = x_0 = 0) \text{ and } \bar{x}_3 \text{ if } (x_2 \text{ OR } x_1 \text{ OR } x_0 = 1)$$

$$\Rightarrow Z_3 = x_3 (\bar{x}_2 \bar{x}_1 \bar{x}_0) + \bar{x}_3 (x_2 + x_1 + x_0)$$

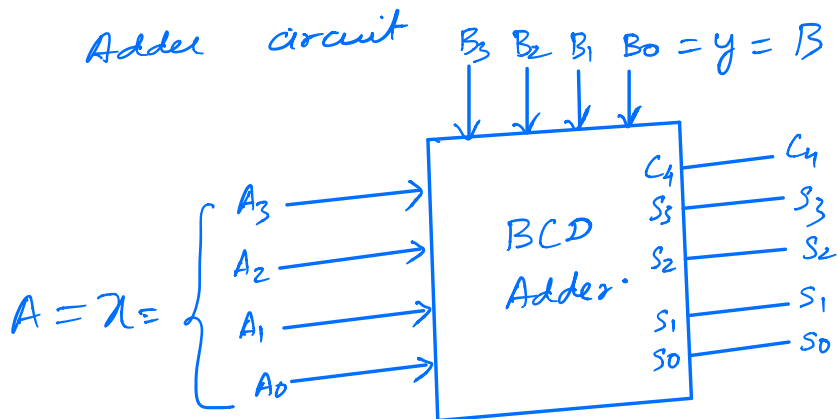
$$= x_3 (\overline{x_2 + x_1 + x_0}) + \bar{x}_3 (x_2 + x_1 + x_0)$$

$$\Rightarrow Z_3 = x_3 \oplus (x_2 + x_1 + x_0).$$

$$C = x_3 x_2 x_1 x_0$$

Design a BCD Adder Subtractor circuit for adding/subtracting two unsigned BCD digits.

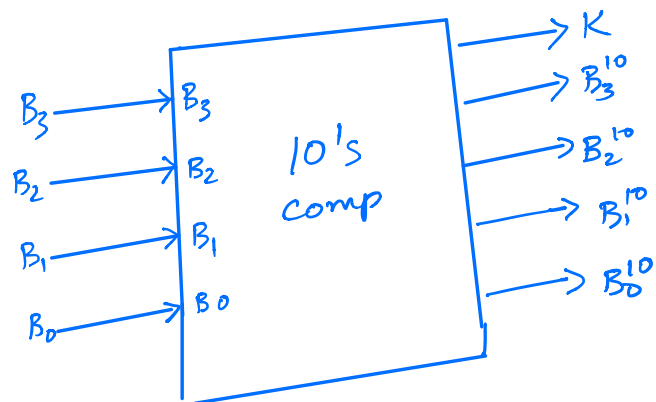
We already have designed/discussed the BCD



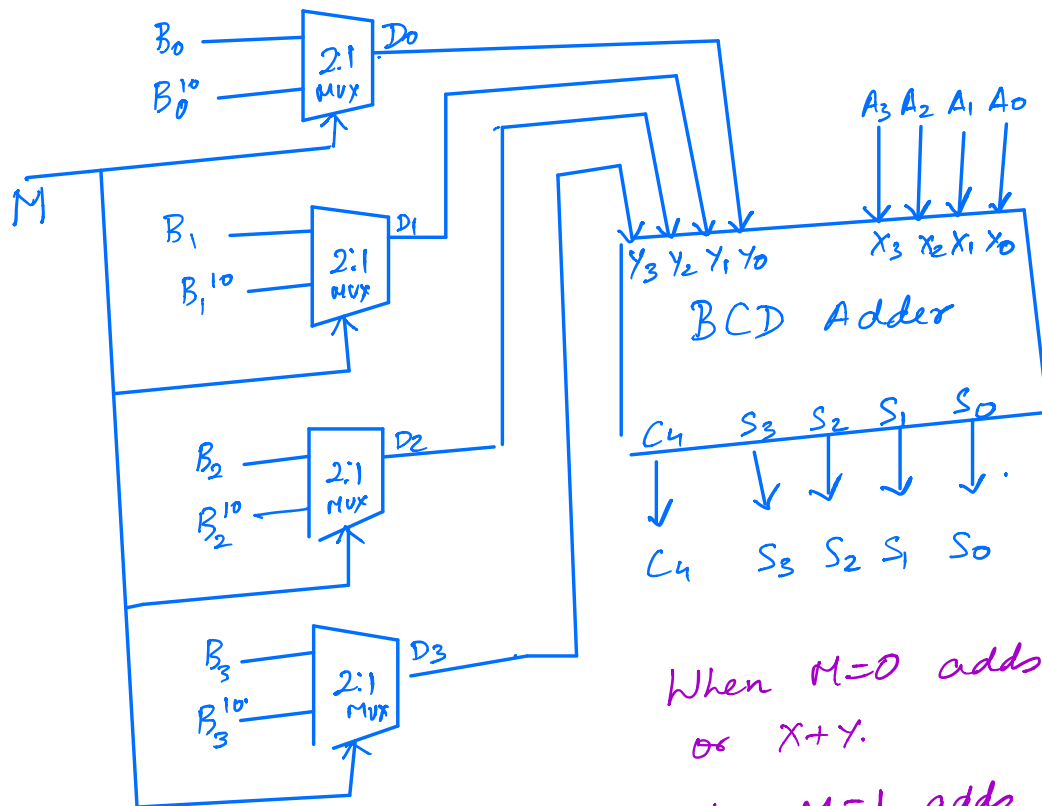
Now we want to add if the condition $M=0$ and add 10's complement of y to x if $M=1$.

Ckt for 10's complement of Y:

B_3	B_2	B_1	B_0	K	B_3^{10}	B_2^{10}	B_1^{10}	B_0^{10}
0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1
0	0	1	0	0	1	0	0	0
⋮								
⋮								
⋮								
⋮								
⋮								
1	0	0	1		0	0	0	1



BCD Adder / 10's complement Adder Ckt:



When $M=0$ adds $A+B$
or $X+Y$.

When $M=1$ adds $A + 10\text{'s comp}$
of B

The o/p in case of 10's complement addition is left as 10's complement.